Small cycles of the Juggler map

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Abstract

The Juggler map is the nonlinear integer map

$$J(n) = \begin{cases} \lfloor \sqrt{n} \rfloor, & n \text{ even}, \\ \lfloor n^{3/2} \rfloor, & n \text{ odd}. \end{cases}$$

It is conjectured that every positive integer eventually reaches 1. We do not prove that conjecture.

The main theorem is a small-cycle census: the Juggler map has no nontrivial cycle of length at most seven. The argument is elementary. A realized parity word w obeys the power envelope $\left(J^{|w|}(n)\right)^{2^{|w|}} \leq n^{3^{\#O(w)}}$, so a cycle word is formally expanding. Inverse cells and two next-square thresholds exclude every even-terminating expanding word of length at most six except two leftover shapes, $OOOEOE$ and $OOOOEE$. Those two are excluded by a tail inequality for $n \geq 256$ and a check of the 254 starts $2 \leq n < 256$. Length seven is the same two-even type: the recorded thresholds and one internal-even bootstrap leave two leftover shapes, $OOOOEOE$ and $OOOOOEE$, excluded by a sharper tail for $n \geq 14$ and a check of the 12 starts $2 \leq n < 14$. Length eight is the first even-terminating expanding length outside that census.

The leftover even-terminating two-even words form two infinite families, $O^{k-2}EE$ and $O^{k-3}EOE$. Both are excluded at every length $k \geq 6$. On a cycle minimum, a three-even leftover with a sufficiently long second gap reduces to those families. Seven bunched three-even families, O^aEEE, O^aEOEE, O^aEOOEE, $O^aEOOOEE$, O^aEEOE, O^aEOEOE, and $O^aEOOEOE$, are likewise excluded. These are family exclusions, not a census at length eight or nine.

The same envelope gives an exact global defect by keeping the floor remainders. Zero defect characterizes the monochrome power towers; concatenation is a two-term power-gap. The defect is not used as a uniform tax — none exists — and is recorded because it makes the envelope identity exact.

Even starts and odd-to-even starts have a uniform one- or two-step descent. The complementary class is the odd-to-odd starts; many of those still descend after a longer word. No density result is stated or used here.

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1. Introduction

The Juggler sequence was introduced by Pickover [1,2]. The one-step map is OEIS A094683 [3]; the stopping-time table is A007320 [4]. The orbit of 3 is $3, 5, 11, 36, 6, 2, 1$. The orbit of 37 already peaks at 24906114455136. Universal arrival at 1 remains the open conjecture of the subject.

The map combines a contracting even branch with an expanding odd branch,

$$E(n) = \lfloor n^{1/2} \rfloor, \qquad O(n) = \lfloor n^{3/2} \rfloor.$$

A word of length k with o odd letters has ideal exponent $3^o/2^k$. Floors are applied after every letter, and a word is available only when the orbit realizes those parities.

The main theorem of the note is the census: there is no nontrivial cycle of period at most six (Theorem 3.6), and none of period at most seven (Theorem 3.8). The tool is the finite-word envelope (Theorem 2.2), together with the inverse cells and two next-square thresholds. After the census, the same cells exclude two infinite leftover families at every expanding length (Theorem 3.12), transport that comparison across a first even letter on a cycle minimum (Theorem 3.13), exclude the seven bunched three-even families (Theorems 3.14–3.20), and upgrade both gapped leftovers from cycle minima to cycle words (Theorem 3.21). The exact defect (Theorems 2.4–2.6) is the same recurrence with remainders kept rather than dropped. It classifies the rigid zero cases and shows that a uniform per-step slack tax is impossible; it is not needed for the length-six exclusion beyond a short alternative to the exponent comparison on mixed words.

Throughout, $\mathbb{N} = \{1, 2, 3, ...\}$. Write J^k for the k-fold iterate. A nonempty realized word w with $J^{|w|}(n) = n$ is a *cycle word*. The unique fixed point is 1; a cycle is *nontrivial* when it contains some $n \geq 2$.

1.1 Related work

Pickover's later exposition is Chapter 45 of [2]. Weisstein [5] records the map, the stopping-time sequences A094670, A094679, A095908, and a verification of arrival at 1 through 10^6. OEIS A094716 [6] records extreme heights, including the start 48443 whose peak has 972 463 digits. Those computations do not bound the period and are not used in the proofs below.

We know of no published exclusion of nontrivial cycles for the exact floor-power map J. Prasad–Prasad [7] estimate excursion and stopping constants for juggler-like maps by a random-walk large-deviation model; those estimates do not apply here. Small-cycle censuses are a standard first layer for Collatz-like maps, surveyed by Lagarias [8,9]. Those results do not transfer: the branches of J are floor powers rather than affine maps (Crandall [10], Matthews–Watts [11]).

Itinerary-class densities are the subject of a companion manuscript [12] and are not used here.

1.2 Verification

The arguments of Sections 2–4 may be read without machine assistance. An independent Lean check of those arguments lives in the repository named in Section 4; the corresponding theorem names are collected in Appendix A. Lemma 3.5 uses a table of 254 six-step evaluations for $2 \leq n < 256$, one table for each of the two leftover words. Lemma 3.7 uses a table of 12 seven-step evaluations for $2 \leq n < 14$. Lemma 3.11 uses the same window to record that no start $2 \leq n < 256$ realizes seven consecutive odd letters. Theorem 3.12 adds two 254-start tables at length eight and one at length nine; Theorem 3.14 reuses the length-nine table for *OOOOOOEEE* below 128; Theorem 3.15 uses tables below 314 and below 16; Theorem 3.16 reuses the 254-start window at prefix lengths $4, 5, 6$; Theorem 3.18 uses the same two windows as Theorem 3.15; Theorems 3.17, 3.19, and 3.20 reuse the 254-start window at the short expanding prefixes of those families. Those tables are finite computations, not a termination proof.

2. Envelope and defect

Let $\mathcal{B} = \{E, O\}$. A finite word $w \in \mathcal{B}^*$ is *realized* at $n \in \mathbb{N}$ when the successive parities of the orbit of n are exactly the letters of w. Write $J^{|w|}(n)$ for the endpoint after those letters, and $\#O(w)$ for the number of odd letters. When w is realized, this endpoint is the $|w|$-fold iterate.

Theorem 2.1 (fixed-word monotonicity). If $n \leq m$ and both realize w, then $J^{|w|}(n) \leq J^{|w|}(m)$.

Proof. Induct on w. The empty word is immediate. The images after a common realized prefix remain ordered, and both current states realize the same next letter. The even branch is the monotone integer square root; the odd branch is $x \mapsto x^3$ followed by the integer square root. □

The realizing set of a fixed word need not be an interval: OE is realized at 7 and at 11 but not at 9.

Theorem 2.2 (finite-word power envelope). If w is realized at n and $m = J^{|w|}(n)$, then

$$m^{2^{|w|}} \leq n^{3^{\#O(w)}}.$$

Proof. The empty word is the equality $n \le n$. Suppose a realized prefix of length ℓ with odd count o ends at x and satisfies $x^{2^\ell} \le n^{3^o}$, and the next letter is realized.

If the next letter is even, then $J(x)^2 \le x$. Raising the inductive bound to the second power gives

$$J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \le x^{2^\ell} \le n^{3^o}.$$

The odd count is unchanged.

If the next letter is odd, then $J(x)^2 \le x^3$, so

$$J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \le x^{3\cdot 2^\ell} = (x^{2^\ell})^3 \le \left(n^{3^o}\right)^3 = n^{3^{o+1}}.$$

This is the claimed bound after one more odd letter. □

Corollary 2.3 (exponent-gap contraction). If $n \ge 2$, w is realized at n, and $3^{\#O(w)} < 2^{|w|}$, then $J^{|w|}(n) < n$.

Proof. Let $m = J^{|w|}(n)$ and $k = |w|$. Theorem 2.2 gives $m^{2^k} \le n^{3^{\#O(w)}}$. The exponent gap and $n \ge 2$ give $n^{3^{\#O(w)}} < n^{2^k}$, so $m^{2^k} < n^{2^k}$. Since $m \ge 1$, one has $m < n$. □

The corollary includes familiar contracting blocks such as *OOOEE*. It does not prove that every start realizes some contracting word.

The floor slack is exact. For a single branch,

$$x^e = J(x)^2 + \rho(x), \qquad e = \begin{cases} 1, & x \text{ even}, \\ 3, & x \text{ odd}, \end{cases}$$

with $0 \le \rho(x) < 2J(x) + 1$. Write

$$\mathrm{gap}(a, \rho, e) = (a+\rho)^e - a^e,$$

so $a^e + \mathrm{gap}(a, \rho, e) = (a+\rho)^e$. If $e \ge 1$, then $\mathrm{gap}(a, \rho, e) = 0$ if and only if $\rho = 0$, and $\rho^e \le \mathrm{gap}(a, \rho, e)$.

Along a realized word w, write $x_0 = n$ and $x_{j+1} = J(x_j)$, and let $\rho_j = \rho(x_j)$. Define a running slack by $D_0 = 0$ and

$$D_{j+1} = \begin{cases} D_j + \mathrm{gap}((x_{j+1})^2, \rho_j, 2^j), & \text{next letter even}, \\ \mathrm{gap}((x_{j+1})^2, \rho_j, 2^j) + \mathrm{gap}(x_j^{2^j}, D_j, 3), & \text{next letter odd}. \end{cases}$$

The *global defect* is $\Delta_w(n) = D_{|w|}$. An even letter keeps the old slack and lifts the new remainder through 2^j. An odd letter first cubes the running slack and then lifts the new remainder. In particular Δ is not the sum of the local remainders.

Theorem 2.4 (global defect identity). If w is realized at n and $m = J^{|w|}(n)$, then

$$n^{3^{\#O(w)}} = m^{2^{|w|}} + \Delta_w(n), \qquad \Delta_w(n) \ge 0.$$

Theorem 2.2 is the inequality $\Delta_w(n) \ge 0$.

Proof. The empty word is $n = n + 0$. Suppose a realized prefix of length ℓ with odd count o ends at x and satisfies $n^{3^o} = x^{2^\ell} + D$.

If the next letter is even, then $x = J(x)^2 + \rho(x)$, so

$$n^{3^o} = (J(x)^2 + \rho(x))^{2^\ell} + D = J(x)^{2^{\ell+1}} + D + \mathrm{gap}(J(x)^2, \rho(x), 2^\ell).$$

The odd count is unchanged, and the new slack is the even update.

If the next letter is odd, then $x^3 = J(x)^2 + \rho(x)$. Cubing the inductive identity gives

$$n^{3^{o+1}} = (x^{2^\ell} + D)^3 = x^{3\cdot 2^\ell} + \mathrm{gap}(x^{2^\ell}, D, 3).$$

The leading term is

$$x^{3\cdot 2^\ell} = (J(x)^2 + \rho(x))^{2^\ell} = J(x)^{2^{\ell+1}} + \mathrm{gap}(J(x)^2, \rho(x), 2^\ell),$$

which is the odd update. □

For a one-letter illustration, take $n = 3$ and $w = O$. Then $J(3) = 5$ and $3^3 = 27 = 5^2 + 2$, so $\Delta_O(3) = 2$. After a later letter the lift is a power-gap, not a sum of remainders.

Theorem 2.5 (vanishing). If w is realized at n, the following are equivalent.

1. $\Delta_w(n) = 0$.
2. Every local remainder along w vanishes.
3. $\left(J^{|w|}(n)\right)^{2^{|w|}} = n^{3^{\#O(w)}}$.

In that case w is monochrome. More precisely, either $w = E^k$ and $n = a^{2^k}$ for an even a, or $w = O^k$ and $n = a^{2^k}$ for an odd a. A realized mixed word therefore has $\Delta_w(n) > 0$.

Proof. The identity of Theorem 2.4 gives (1)⇔(3). For (1)⇔(2), use the recurrence. If $e \geq 1$, a power-gap vanishes if and only if its addend vanishes. The even update is a sum of two nonnegative terms, and the odd update is a sum of two power-gaps; either vanishes only when the incoming slack and the new remainder both vanish. Thus $D_{|w|} = 0$ forces $D_0 = 0$ and every $\rho_j = 0$. The converse is immediate.

If every remainder vanishes, an even step satisfies $x_j = x_{j+1}^2$, while an odd step satisfies $x_j^3 = x_{j+1}^2$. In either case x_j and x_{j+1} have the same parity, so the itinerary is monochrome. If $w = E^k$, repeated substitution gives $n = x_k^{2^k}$; writing $a = x_k$, its parity is even. If $w = O^k$, unique factorization applied to $x_j^3 = x_{j+1}^2$ shows that every prime valuation of x_j is even. Iterating this valuation relation shows that every prime valuation of $n = x_0$ is divisible by 2^k, hence $n = a^{2^k}$ for an odd a. Conversely, these even and odd towers make every local remainder zero. □

Theorem 2.6 (composition). If u is realized at n and v is realized at $m = J^{|u|}(n)$, then

$$\Delta_{uv}(n) = \mathrm{gap}(m^{2^{|u|}},\, \Delta_u(n),\, 3^{\#O(v)}) + \mathrm{gap}(\left(J^{|v|}(m)\right)^{2^{|v|}},\, \Delta_v(m),\, 2^{|u|}).$$

In particular $\Delta_v(m)^{2^{|u|}} \leq \Delta_{uv}(n)$. Every local remainder satisfies $\rho_j^{2^j} \leq \Delta_w(n)$.

Proof. Write $o_u = \#O(u)$ and $o_v = \#O(v)$. Theorem 2.4 on u and on uv gives

$$n^{3^{o_u+o_v}} = \left(m^{2^{|u|}} + \Delta_u(n)\right)^{3^{o_v}} = \left(J^{|uv|}(n)\right)^{2^{|uv|}} + \Delta_{uv}(n).$$

The first power expands as

$$m^{2^{|u|}\cdot 3^{o_v}} + \mathrm{gap}(m^{2^{|u|}},\, \Delta_u(n),\, 3^{o_v}).$$

Theorem 2.4 on v at m rewrites the leading term as

$$\left(\left(J^{|v|}(m)\right)^{2^{|v|}} + \Delta_v(m)\right)^{2^{|u|}},$$

which expands as the second gap plus $\left(J^{|uv|}(n)\right)^{2^{|uv|}}$. Comparing the two expressions for $n^{3^{o_u+o_v}}$ yields the identity. The suffix inequality is $\rho^e \leq \mathrm{gap}(a, \rho, e)$ on the second summand. For a local remainder at index j, split w after j letters: the suffix defect is at least ρ_j, and raising through 2^j gives the stated bound. □

Composition is polynomial, not additive: later odd letters raise the prefix slack to the third power. The naive recurrence $\Delta \leftarrow \Delta + \rho$ is false.

Corollary 2.7 (cycle surplus). If w is a cycle word at n, then

$$\Delta_w(n) = n^{3^{\#O(w)}} - n^{2^{|w|}}$$

exactly.

Proof. Theorem 2.4 with $m = n$. $\square$

The identity does not supply a state-independent positive tax, and none exists. For one realized letter at x with branch exponent $e \in \{1, 3\}$,

$$x^e < (J(x) + 1)^2,$$

so the relative slack $1 + \eta = x^e/J(x)^2$ satisfies

$$\eta < \frac{2}{J(x)} + \frac{1}{J(x)^2},$$

which tends to 0 as the state grows. The recorded extreme is an *OOE* block at $n = 180370579261640036336071806107777 \approx 1.80 \cdot 10^{32}$ whose word-relative slack

$$q_w(n) = \frac{n^{3^{\#O(w)}}}{\left(J^{|w|}(n)\right)^{2^{|w|}}} - 1$$

satisfies $0 < q_{OOE}(n) < 10^{-30}$, by the exact integer comparison $0 < (n^9 - J^3(n)^8)\, 10^{30} < J^3(n)^8$.

3. Inverse cells and the census

The exact one-step fibers are

$$J(n) = q \iff q^2 \le n < (q+1)^2 \quad (n \text{ even})$$

and

$$J(n) = m \iff m^2 \le n^3 < (m+1)^2 \quad (n \text{ odd}).$$

Lemma 3.1 (odd cells are unique). An odd fiber contains at most one integer. An even fiber is a parity-restricted square interval and may contain many predecessors.

Proof. Suppose $a < b$ lie in the same odd cell indexed by m. Then $(a+1)^3 \le b^3 < (m+1)^2$ and $m^2 \le a^3$. Subtracting the latter lower bound from the former upper bound gives

$$3a^2 + 3a + 1 = (a+1)^3 - a^3 < (m+1)^2 - m^2 = 2m + 1,$$

hence $3a^2 + 3a < 2m$ and $3a^2 < 2m$. If $a = 0$, then $m = 0$ and the displayed strict inequality is already impossible. If $a > 0$, squaring and using $m^2 \le a^3$ gives $9a^4 < 4m^2 \le 4a^3$, contradicting $4a^3 < 9a^4$. $\square$

Theorem 3.2 (cycle restrictions). Let w be a cycle word at $n \ge 2$.

(i) The word is formally expanding:

$$2^{|w|} < 3^{\#O(w)}.$$

A contracting word cannot close a nontrivial cycle.

(ii) The cycle minimum is odd and the cycle maximum is even. A minimum-based orientation cannot end in an odd letter.

(iii) A realized word v is *superquadratic* if

$$3^{\#O(v)} \ge 2^{|v|+1}.$$

Any realized path from a start $n \ge 2$ to a state at least n^2 is superquadratic. On a cycle minimum the path to any later even state is superquadratic. The prefix *OOE* is expanding ($9 > 8$) but not superquadratic ($9 < 16$), so it cannot carry the minimum to square scale.

Proof. For (i), Theorem 2.2 applied to a cycle endpoint gives $n^{2^{|w|}} \le n^{3^{\#O(w)}}$. Since $n \ge 2$, comparison of exponents gives $2^{|w|} \le 3^{\#O(w)}$; equality is impossible because the two sides have different prime divisors and $|w| \ge 1$. Alternatively: a mixed cycle word has $\Delta_w(n) > 0$ by Theorem 2.5, so the envelope is strict; a monochrome tower cannot return for $n \ge 2$.

Every state on such a cycle is at least 2: once an orbit reaches 1, it remains there and cannot return to a start $n \ge 2$. An even state $x \ge 2$ satisfies $J(x) < x$, so a cycle minimum cannot be even. An odd state

$x \geq 3$ satisfies $J(x) > x$, so a cycle maximum cannot be odd. This proves the first assertion of (ii). For the final-letter assertion, let x be the predecessor of a minimum-oriented return n. If the last letter were odd, the odd return cell would give $n^2 \leq x^3 < (n+1)^2$. Minimality gives $x \geq n$, hence $n^3 < (n+1)^2$, impossible for $n \geq 3$; and the minimum is odd, so $n \geq 3$.

For (iii), let a realized word v send n to $y \geq n^2$. Theorem 2.2 gives

$$n^{2^{|v|+1}} = (n^2)^{2^{|v|}} \leq y^{2^{|v|}} \leq n^{3^{\#O(v)}}.$$

Since $n \geq 2$, comparison of exponents proves the superquadratic inequality. On a cycle minimum, if a later state y is even, its successor is at least n. Thus $n \leq J(y) = \lfloor \sqrt{y} \rfloor$, so $n^2 \leq y$, and the preceding argument applies to that prefix. □

Lemma 3.3 (coarse lower envelope). For $n \geq 1$, write $q = \lfloor \sqrt{n} \rfloor$. Then $q^2 \leq n < (q+1)^2$. For $q \geq 1$ one has $(q+1)^2 \leq 4q^2$, and the second inequality is strict for $q \geq 2$, while for $q = 1$ one has $n < 4$. In all cases

$$n < 4 \lfloor \sqrt{n} \rfloor^2.$$

Thus an even step satisfies $n \leq 4\, J(n)^2$ and an odd step satisfies $n^3 \leq 4\, J(n)^2$. Composing along a realized word v of length k with odd count o gives

$$n^{3^o} \leq C_v\, J^k(n)^{2^k}.$$

The constant starts at 1 and updates by $C \mapsto C \cdot 4^{2^j}$ on an even letter and $C \mapsto C^3 \cdot 4^{2^j}$ on an odd letter, at step j. In particular $C_{OOO} = 2^{38}$ and $C_{OOOO} = 2^{130}$.

Proof. The comparison $n < 4q^2$ is the paragraph above. The composition law is the same recurrence as the proof of Theorem 2.2, with a factor 4^{2^j} inserted at each letter. The values C_{OOO} and C_{OOOO} are the result of that recurrence on those two words. □

Lemma 3.4 (next-square thresholds). (i) If $q \geq 5$ realizes OO, then $J^2(q) \geq (q+1)^2$. (ii) If $q \geq 3$ realizes OOO, then $J^3(q) \geq (q+1)^2$. (iii) If a realized word v satisfies $J^{|v|}(q) \geq (q+1)^2$ and the next realized letter is odd, then $J^{|v|+1}(q) \geq (q+1)^2$. (iv) If vE is a cycle word at n, then $J^{|v|}(n) < (n+1)^2$. (v) If $a \geq 3$, then O^aE is not a cycle word at any $n \geq 2$.

Proof. For (i), write $m = \lfloor q^{3/2} \rfloor$. The bound $J^2(q) \geq (q+1)^2$ follows from $m^3 \geq (q+1)^4$, because then $\lfloor m^{3/2} \rfloor \geq (q+1)^2$. If $q = 5$, then $11^2 = 121 \leq 125 = 5^3 < 144 = 12^2$, so $m = 11$, and $11^3 = 1331 \geq 6^4 = 1296$. Since q realizes OO, it is odd, so the only remaining case is $q \geq 7$. Then $m \geq q^{3/2} - 1$, so it is enough that $(q^{3/2}-1)^3 \geq (q+1)^4$. Expanding the left side and dropping the positive remainder $3q^{3/2} - 1$ reduces this to $q^{9/2} - 3q^3 \geq (q+1)^4$, or equivalently $\sqrt{q} - 3/q \geq (1+1/q)^4$. The left side increases for $q > 0$ and the right side decreases, so the case $q = 7$ suffices:

$$\sqrt{7} - \frac{3}{7} > \frac{5}{2} - \frac{3}{7} = \frac{29}{14} > \frac{4096}{2401} = \left(\frac{8}{7}\right)^4,$$

where $\sqrt{7} > 5/2$ follows from $25/4 < 7$.

For (ii), the orbit $3 \to 5 \to 11 \to 36$ gives $J^3(3) = 36 \geq 16$. If $q \geq 5$ realizes OOO, then it realizes OO, so (i) gives $J^2(q) \geq (q+1)^2$. The third letter is odd, hence $J^3(q) = J(J^2(q)) \geq J^2(q)$.

For (iii), the image after v is odd and at least $(q+1)^2 \geq 4$, so the odd branch does not decrease it.

For (iv), the last letter is even, so the preimage $z = J^{|v|}(n)$ is even and satisfies $n^2 \leq z < (n+1)^2$.

For (v), suppose O^aE is a cycle word at $n \geq 2$. The start is odd, hence at least 3, and realizes O^a. Parts (ii) and (iii) give $J^a(n) \geq (n+1)^2$, contradicting (iv). □

Lemma 3.5 (two length-six exclusions). Neither $OOOEOE$ nor $OOOOEE$ is a cycle word at any $n \geq 2$.

Proof. First, if $n \geq 256$, then

$$n^{81} > 2^{130}(n+1)^{64}.$$

Indeed $257^{64} < 2{\cdot}256^{64}$ because $(1{+}1/256)^{64} < e^{64/256} = e^{1/4} < 2$, and $256(n{+}1) \le 257\,n$, so $(n{+}1)^{64} < 2n^{64}$. Then $2^{130}(n{+}1)^{64} < 2^{131}n^{64}$. Since $n \ge 256 = 2^8$, one has $n^{17} \ge 2^{136}$, and therefore $2^{131}n^{64} < 2^{136}n^{64} \le n^{81}$.

For $2 \le n < 256$, neither word returns to its start. This is a table of 254 evaluations of each word: at every start, either some letter fails to match the current parity, or the six-step image differs from the start. The same finite check is the Lean `native_decide` evaluation behind `no_cycle_word_oooeoe` and `no_cycle_word_ooooee` (Appendix A).

Now suppose $OOOOEE$ is a cycle word at $n \ge 256$, and write $z = J^4(n)$ for the image after the prefix $OOOO$. Lemma 3.4(iv) gives $J(z) < (n+1)^2$, and the preceding even letter gives $z < (J(z)+1)^2$, hence $z < (n+1)^4$. Lemma 3.3 on $OOOO$ gives $n^{81} \le 2^{130}z^{16}$, so $n^{81} < 2^{130}(n+1)^{64}$, contradicting the tail inequality.

Finally suppose $OOOEOE$ is a cycle word at $n \ge 256$. Write $z_3 = J^3(n)$ and $y = J(z_3) = \lfloor\sqrt{z_3}\rfloor$, so $z_3 < (y+1)^2$. Lemma 3.3 on OOO gives $n^{27} \le 2^{38}z_3^8 < 2^{38}(y+1)^{16}$. Cubing yields $n^{81} < 2^{114}(y+1)^{48}$. The last letters OE give the odd-cell bound $y^3 < (n+1)^4$. Write $A = n+1 \ge 257$. We claim $(y+1)^3 < 2A^4$. If $y \le A$, this is $(A+1)^3 < 2A^4$. If $y > A$, then $y \ge 258$, so $3y+1 < y^2$ and hence $(y+1)^3 = y^3+3y^2+3y+1 < A^4 + 4y^2$, while $4y^2 < A^4$ because $4y^3 < 4A^4 \le yA^4$. Raising $(y+1)^3 < 2A^4$ to the sixteenth power gives $(y+1)^{48} < 2^{16}(n+1)^{64}$. Combining with the cubed lower envelope produces again $n^{81} < 2^{130}(n+1)^{64}$. □

Theorem 3.6 (small-cycle census). No word of length at most six is a cycle word at any $n \ge 2$. Equivalently, a nontrivial Juggler cycle, if one exists, has period at least seven.

Proof. Rotating a cycle word by one letter moves the base point one step along the orbit and yields another cycle word; every state of the cycle is at least 2, because an orbit that reaches 1 stays at 1 and cannot return to a start $n \ge 2$.

If every letter is odd, the start is odd, hence $n \ge 3$, and the odd branch strictly increases there: $J(x) > x$ for odd $x \ge 3$, since $x^3 \ge (x+1)^2$. The orbit ascends strictly and never returns. Otherwise some letter is even, and a rotation ending just after that letter produces an even-terminating cycle word vE of the same length based at a cycle state $m \ge 2$. It therefore suffices to exclude even-terminating cycle words of length at most six.

By Theorem 3.2(i) a cycle word is formally expanding. No even-terminating word of length one or two is expanding ($2 > 3^0$ and $4 > 3$).

Length three: the only expanding candidate is OOE. If $m \ge 5$ realizes OO, Lemma 3.4(i) gives $J^2(m) \ge (m+1)^2$, contradicting Lemma 3.4(iv). The only smaller odd start is $m = 3$, where $J^2(3) = 11$ is odd, so the final even letter is not realized.

Length four: the expanding filter requires three odd letters among the first three, leaving only O^3E, excluded by Lemma 3.4(v).

Length five: the filter requires four odd letters among the first four, leaving only O^4E, excluded by Lemma 3.4(v).

Length six: the filter requires at least four odd letters among the first five, leaving O^5E, $EOOOOE$, $OEOOOE$, $OOEOOE$, $OOOEOE$, and $OOOOEE$. Lemma 3.4(v) excludes O^5E. The word $EOOOOE$ rotates one step onto $OOOOEE$, and $OEOOOE$ rotates two steps onto $OOOEOE$; both are excluded by Lemma 3.5.

It remains to exclude $OOEOOE$. Let this word be a cycle word at some start, and rotate to a cycle minimum $m \ge 2$. The three alignments of the two even letters are $OOEOOE$, $OEOOEO$, and $EOOEOO$. The last starts even, so it cannot be a minimum, by Theorem 3.2(ii). The middle starts OE: the first image is even and strictly below m^2, whereas every even state after a cycle minimum is at least m^2, as proved in Theorem 3.2(iii). Thus the minimum orientation is $OOEOOE$. In particular m is odd, so $m \ge 3$. The prefix OOE must be realized. If $m = 3$, then $3 \to 5 \to 11$ and 11 is odd, so OOE is not realized. Hence $m \ge 5$. Write $y = J^3(m)$ for the state after OOE. Then $y \ge m$ by minimality, so $y \ge 5$. The suffix OO is realized at y, and Lemma 3.4(i) gives $J^2(y) \ge (y+1)^2 \ge (m+1)^2$. The last letter is even, so Lemma 3.4(iv) gives $J^2(y) < (m+1)^2$. □

Lemma 3.4(v) excludes every odd-run-then-even word O^aE with $a \geq 3$, of any length.

Lemma 3.7 (two length-seven exclusions). Neither *OOOOEOE* nor *OOOOOEE* is a cycle word at any $n \geq 2$.

Proof. First, if $n \geq 14$, then

$$n^{243} > 2^{422}(n+1)^{128}.$$

Indeed $14(n+1) \leq 15n$, so $(n+1)^{128} \leq (15/14)^{128} n^{128}$. The finite comparison $2^{422}15^{128} < 14^{243}$ then yields $2^{422}(n+1)^{128} < 14^{115} n^{128} \leq n^{243}$.

For $2 \leq n < 14$, neither word is realized: at every such start, some letter fails to match the current parity. The same finite check is the Lean `native_decide` evaluation behind `no_cycle_word_ooooeoe` and `no_cycle_word_oooooee` (Appendix A).

Now suppose *OOOOOEE* is a cycle word at $n \geq 14$, and write $z = J^5(n)$ for the image after the prefix *OOOOO*. Lemma 3.4(iv) on the last even letter, together with the preceding even letter, gives $z < (n+1)^4$. Lemma 3.3 on *OOOOO* gives $n^{243} \leq 2^{422} z^{32}$, so $n^{243} < 2^{422}(n+1)^{128}$, contradicting the tail inequality.

Finally suppose *OOOOEOE* is a cycle word at $n \geq 14$. Write $z_4 = J^4(n)$ and $y = J(z_4) = \lfloor \sqrt{z_4} \rfloor$, so $z_4 < (y+1)^2$. Lemma 3.3 on *OOOO* gives $n^{81} \leq 2^{130} z_4^{16} < 2^{130}(y+1)^{32}$. Cubing yields $n^{243} < 2^{390}(y+1)^{96}$. The last letters *OE* give the odd-cell bound $y^3 < (n+1)^4$. Write $A = n+1 \geq 15$. The same comparison $(y+1)^3 < 2A^4$ as in Lemma 3.5 holds at this smaller scale. Raising to the thirty-second power gives $(y+1)^{96} < 2^{32}(n+1)^{128}$. Combining with the cubed lower envelope produces again $n^{243} < 2^{422}(n+1)^{128}$. □

Theorem 3.8 (small-cycle census through length seven). No word of length at most seven is a cycle word at any $n \geq 2$. Equivalently, a nontrivial Juggler cycle, if one exists, has period at least eight.

Proof. The reduction of Theorem 3.6 applies at every length: an all-odd word cannot return, and every mixed cycle word rotates to an even-terminating cycle word based at a cycle state $m \geq 2$. Lengths at most six are Theorem 3.6. It remains to exclude even-terminating cycle words of length seven.

By Theorem 3.2(i) a cycle word is formally expanding. The even-terminating expanding length-seven words are exactly O^6E, EO^5E, OEO^4E, $OOEO^3E$, O^3EO^2E, O^4EOE, and O^5EE. Lemma 3.4(v) excludes O^6E. The word EO^5E rotates one step onto O^5EE, and OEO^4E starts OE, so it cannot be a cycle minimum (Theorem 3.2(iii)) and rotates two steps onto O^4EOE; both leftovers are excluded by Lemma 3.7.

It remains to exclude $OOEO^3E$ and O^3EO^2E. Rotate either word to a cycle minimum $m \geq 2$. For $OOEO^3E$ the minimum orientation retains the internal even letter followed by the suffix *OOO*. Then $m \geq 3$, the prefix through that even letter is realized, and Lemma 3.4(ii) at threshold 3 contradicts the last even cell. For O^3EO^2E the same bootstrap applies with suffix *OO* and threshold 5, once $m = 3$ is removed: at $m = 3$ the state after *OOO* is even, so the next even letter is not realized. □

The census of Theorems 3.6 and 3.8 is a statement about period. The same cells organise leftover words by even-count. Write

$$e_a = 2(3^a - 2^a)$$

for $a \geq 0$.

Lemma 3.9 (trailing even run). If a cycle word based at n ends with $r \geq 1$ even letters, the state immediately before that run is strictly less than $(n+1)^{2^r}$.

Proof. The case $r = 1$ is Lemma 3.4(iv). Suppose the claim holds for some $r \geq 1$, and let vE^{r+1} be a cycle word at n. Write $z = J^{|v|}(n)$. The inductive hypothesis applied to the suffix E^r after the first of those even letters gives $J(z) < (n+1)^{2^r}$. The state z is even, so $z < (J(z)+1)^2 \leq \left((n+1)^{2^r}\right)^2 = (n+1)^{2^{r+1}}$. □

Lemma 3.10 (odd-run lower envelope). If $n \geq 1$ realizes O^a, then

$$n^{3^a} \leq 2^{e_a} J^a(n)^{2^a}.$$

Proof. Lemma 3.3 supplies a multiplicative constant C_v along any realized word, with $C_\varepsilon = 1$, $C \mapsto C \cdot 4^{2^j}$ on an even letter, and $C \mapsto C^3 \cdot 4^{2^j}$ on an odd letter, at step j. On the pure odd word O^a every letter is odd, so $C_{O^{a+1}} = C^3_{O^a} \cdot 4^{2^a}$. Writing $C_{O^a} = 2^{e_a}$ with $e_0 = 0$ yields the recurrence $e_{a+1} = 3e_a + 2^{a+1}$, because $4^{2^a} = 2^{2^{a+1}}$. The closed form $e_a = 2(3^a - 2^a)$ satisfies the recurrence and the initial value. □

Lemma 3.11 (seven-odd window). No integer n with $2 \le n < 256$ realizes the word O^7.

Proof. This is a table of 254 seven-step evaluations: at every such start, some letter fails to match the current parity. The same finite check is the Lean `native_decide` evaluation behind `no_follows_seven_odds_of_lt256` (Appendix A). □

Theorem 3.12 (two-even leftover families). Let $k \ge 6$ and $n \ge 2$. Neither $O^{k-2}EE$ nor $O^{k-3}EOE$ is a cycle word at n.

Proof. First, if $n \ge 256$, then
$$n^{3^{k-2}} > 2^{e_{k-2}}(n+1)^{2^k}.$$
The case $k = 6$ is the tail inequality of Lemma 3.5. If the display holds at some $k \ge 6$, cubing both sides produces
$$n^{3^{k-1}} > 2^{3e_{k-2}}(n+1)^{3 \cdot 2^k}.$$
The recurrence of Lemma 3.10 gives $e_{k-1} = 3e_{k-2} + 2^{k-1}$, so the desired comparison at length $k+1$ reduces to $2^{2^{k-1}} < (n+1)^{2^k}$. Equivalently $2 < (n+1)^2$, which holds for every $n \ge 256$.

Now suppose $O^{k-2}EE$ is a cycle word at such an n. Write $z = J^{k-2}(n)$. Lemma 3.9 with $r = 2$ gives $z < (n+1)^4$. Lemma 3.10 on the prefix O^{k-2} gives $n^{3^{k-2}} \le 2^{e_{k-2}} z^{2^{k-2}}$, hence $n^{3^{k-2}} < 2^{e_{k-2}}(n+1)^{2^k}$, contradicting the tail.

Finally suppose $O^{k-3}EOE$ is a cycle word at such an n. Write $z = J^{k-3}(n)$ and $y = \lfloor\sqrt{z}\rfloor$, so $z < (y+1)^2$. Lemma 3.10 on O^{k-3} and cubing produce $n^{3^{k-2}} < 2^{3e_{k-3}}(y+1)^{3 \cdot 2^{k-2}}$. The last letters OE give the odd-cell bound $y^3 < (n+1)^4$. The comparison $(y+1)^3 < 2(n+1)^4$ of Lemma 3.5 applies at this scale. Raising it to the power 2^{k-2} and using $e_{k-2} = 3e_{k-3} + 2^{k-2}$ recovers again $n^{3^{k-2}} < 2^{e_{k-2}}(n+1)^{2^k}$.

For $2 \le n < 256$, the cases $k = 6$ and $k = 7$ are Lemmas 3.5 and 3.7. The remaining short words O^6EE, O^5EOE, and O^6EOE fail to return on the same 254-start window; this is the Lean `native_decide` evaluation behind `no_cycle_word_two_even_ee` and `no_cycle_word_two_even_eoe` (Appendix A). Any longer leftover of either family begins with seven consecutive odd letters, which Lemma 3.11 forbids on this window. □

A cycle word w at n is *minimum-based* when n is a cycle minimum: $J^j(n) \ge n$ for every $0 \le j < |w|$. The remainder after a proper prefix of a cycle word need not itself be a cycle word at the prefix endpoint. The next statement therefore transports the tail inequality of Theorem 3.12, not the cycle-word exclusion at a later start.

Theorem 3.13 (first-even transport). Let $n \ge 2$. No minimum-based cycle word at n has the form O^aEO^bEE with $a \ge 2$ and $b \ge 4$, or the form O^aEO^bEOE with $a \ge 2$ and $b \ge 3$.

Proof. Write $y = J^{a+1}(n)$ for the state after the first even letter. Minimum-basedness gives $y \ge n$. In the first family the remainder after that letter is O^bEE with $b+2 \ge 6$; in the second it is O^bEOE with $b+3 \ge 6$. The trailing-even and last-odd cells of those remainders are measured against the cycle start n. Combined with Lemma 3.10 at the remainder start y, the same algebra as in Theorem 3.12 produces
$$y^{3^{\ell-2}} < 2^{e_{\ell-2}}(n+1)^{2^\ell} \le 2^{e_{\ell-2}}(y+1)^{2^\ell},$$
where ℓ is the remainder length. If $y \ge 256$, the first paragraph of Theorem 3.12 supplies the opposite inequality at y.

If $y < 256$, then $n \le y < 256$. A gapped leftover of total length at least 17 has $a \ge 7$ or $b \ge 7$, so either the prefix or the remainder realizes seven consecutive odd letters, contradicting Lemma 3.11. The finitely many short-gap words with $2 \le a \le 6$ and $b \le 6$ fail to be minimum-based cycle words on the window

$2 \le n < 256$; this is the Lean `native_decide` evaluation behind `no_cycleMin_gapped_three_even_ee` and `no_cycleMin_gapped_three_even_eoe` (Appendix A). $\square$

The hypothesis that the start is a cycle minimum is essential. If $y < n$, the leftover cell is measured against a larger start and need not contradict the tail at y. In particular, Theorem 3.13 does not assert that those words fail to be cycle words at a non-minimum start. That upgrade is Theorem 3.21.

Theorem 3.14 (three trailing evens). Let $a \ge 6$ and $n \ge 2$. The word O^aEEE is not a cycle word at n.

Proof. Write $z = J^a(n)$. Lemma 3.9 with $r = 3$ gives $z < (n+1)^8$. Lemma 3.10 then yields $n^{3^a} < 2^{e_a}(n+1)^{2^{a+3}}$ on any such cycle word. For $n \ge 128$ the opposite comparison

$$n^{3^a} > 2^{e_a}(n+1)^{2^{a+3}}$$

holds. The case $a = 6$ is $n^{729} > 2^{1330}(n+1)^{512}$. Indeed, for $n \ge 128$ one has $(n+1)^{512} < (129/128)^{512} n^{512} < e^4 n^{512} < 64\, n^{512}$, so the claimed bound reduces to $n^{217} > 2^{1336}$. Since $n \ge 128 = 2^7$, the left side is at least 2^{1519}. Cubing the $a = 6$ comparison produces the general case: the recurrence of Lemma 3.10 reduces the inductive step to $(n+1)^4 > 2$, which holds for every $n \ge 128$.

For $2 \le n < 128$, the case $a = 6$ is a table of evaluations of $OOOOOOEEE$: at every such start the word fails to return. The same finite check is the Lean `native_decide` evaluation behind `no_cycle_word_ooooooeee` (Appendix A). For $a \ge 7$ the prefix contains seven consecutive odd letters, which Lemma 3.11 forbids on this window. $\square$

Theorem 3.15 (mixed bunched family $EOEE$). Let $a \ge 5$ and $n \ge 2$. The word O^aEOEE is not a cycle word at n.

Proof. First let $n \ge 4$, and write $z = J^a(n)$, $y = \lfloor \sqrt{z} \rfloor$, and $p = J(y)$. Lemma 3.9 with $r = 2$ after the prefix O^aEO gives $p < (n+1)^4$. The letter after y is odd, so Lemma 3.10 at length one yields $y^3 \le 4p^2 < 4(n+1)^8$. For $n \ge 4$ one has $4(n+1)^8 < (n+1)^9$, hence $y < (n+1)^3$. The even cell at z then gives $z < (y+1)^2 \le (n+1)^6$. Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{6 \cdot 2^a}.$$

For $a = 5$ and $n \ge 314$, the opposite comparison $n^{243} > 2^{422}(n+1)^{192}$ holds. The base instance is the finite inequality $2^{422}315^{192} < 314^{243}$. If the display holds at some $n \ge 1$, the elementary comparison $n(n+2) < (n+1)^2$ upgrades it to the same display at $n+1$. Cubing then produces the comparison at $a+1$, once $(n+1)^6 > 4$. In particular the case $a = 6$ already holds for every $n \ge 16$.

For $2 \le n < 314$ and $a = 5$, and for $2 \le n < 16$ and $a = 6$, the word fails to return; these are the Lean `native_decide` evaluations behind `no_cycle_word_three_even_eoee` (Appendix A). For $a \ge 7$ and $n < 256$, Lemma 3.11 applies. For $a \ge 6$ and $n \ge 16$, the tail of the previous paragraph applies. $\square$

Theorem 3.16 (mixed bunched family $EOOEE$). Let $a \ge 4$ and $n \ge 2$. The word O^aEOOEE is not a cycle word at n.

Proof. First let $n \ge 32$, and write $z = J^a(n)$, $y = \lfloor \sqrt{z} \rfloor$, and p for the image after the prefix O^aEOO. Lemma 3.9 with $r = 2$ gives $p < (n+1)^4$. The two letters after y are odd, so Lemma 3.10 at length two yields $y^9 \le 2^{10}p^4 < 2^{10}(n+1)^{16}$. For $n \ge 32$ one has $2^{10} < (n+1)^2$, hence $y < (n+1)^2$. The even cell at z then gives $z < (y+1)^2 \le (n+1)^4$. Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{2^{a+2}}.$$

For $n \ge 256$ and $a \ge 4$, this is the opposite of the shared tail of Theorem 3.12 at length $k = a + 2$.

For $2 \le n < 256$, the cases $a = 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eooee` (Appendix A). For $a \ge 7$, Lemma 3.11 applies. $\square$

Theorem 3.17 (mixed bunched family $EOOOEE$). Let $a \ge 3$ and $n \ge 2$. The word $O^aEOOOEE$ is not a cycle word at n.

Proof. First let $a \geq 4$ and $n \geq 3$, and write $z = J^a(n)$, $y = \lfloor \sqrt{z} \rfloor$, and p for the image after the prefix O^aEOOO. Lemma 3.9 with $r = 2$ gives $p < (n+1)^4$. The three letters after y are odd, so Lemma 3.10 at length three yields $y^{27} \leq 2^{38}p^8 < 2^{38}(n+1)^{32}$. For $n \geq 3$ one has $2^{38} < (n+1)^{22}$, hence $y < (n+1)^2$. The even cell at z then gives $z < (y+1)^2 \leq (n+1)^4$. Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{2^{a+2}}.$$

For $n \geq 256$ and $a \geq 4$, this is the opposite of the shared tail of Theorem 3.12 at length $k = a + 2$, already used in Theorem 3.16.

Now let $a = 3$ and $n \geq 256$. The same two-even cell gives $z < (y+1)^2$, so Lemma 3.10 at the prefix O^3 yields $n^{27} < 2^{38}(y+1)^{16}$. The three-odd envelope on y still gives $y^{27} < 2^{38}(n+1)^{32}$.

If $y < 39$, then $n^{27} < 2^{38} \cdot 39^{16}$. The numerical comparison $2^{38} \cdot 39^{16} < 24^{27}$ contradicts $n \geq 24$.

If $y \geq 39$, the numerical comparison $40^{27} < 2{\cdot}39^{27}$ upgrades to $(y+1)^{27} < 2y^{27}$, hence $(y+1)^{27} < 2^{39}(n+1)^{32}$. Cubing the prefix bound three times produces $n^{729} < 2^{1026}(y+1)^{432}$. Raising the successor bound to the sixteenth power produces $(y+1)^{432} < 2^{624}(n+1)^{512}$. Combining these displays yields $n^{729} < 2^{1650}(n+1)^{512}$. The opposite comparison holds at $n = 197$ and persists to every larger start by the elementary comparison $n(n+2) < (n+1)^2$ already used in Theorem 3.15.

For $2 \leq n < 256$, the cases $a = 3, 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eoooee` (Appendix A). For $a \geq 7$, Lemma 3.11 applies. ☐

Theorem 3.18 (mixed bunched family ***EEOE*****).** Let $a \geq 5$ and $n \geq 2$. The word O^aEEOE is not a cycle word at n.

Proof. First let $n \geq 4$, and write $z = J^a(n)$ and y for the last odd letter of O^aEEOE. The suffix EOE is a cycle suffix, so the last-odd cube of Theorem 3.15 gives $y^3 < (n+1)^4$. The two letters between z and y are even, so $z < (y+1)^4$. For $n \geq 4$ the successor comparison $(y+1)^3 < 2(n+1)^4$ upgrades this to $z < (n+1)^6$. Combined with Lemma 3.10, any such cycle word would satisfy the same display as Theorem 3.15:

$$n^{3^a} < 2^{e_a}(n+1)^{6 \cdot 2^a}.$$

The opposite comparison is therefore the tail of Theorem 3.15: it holds for $a = 5$ and $n \geq 314$, and already for $a = 6$ and $n \geq 16$.

For $2 \leq n < 314$ and $a = 5$, and for $2 \leq n < 16$ and $a = 6$, the word fails to return; these are the Lean `native_decide` evaluations behind `no_cycle_word_three_even_eeoe` (Appendix A). For $a \geq 7$ and $n < 256$, Lemma 3.11 applies. For $a \geq 6$ and $n \geq 16$, the tail of the previous paragraph applies. ☐

Theorem 3.19 (mixed bunched family ***EOEOE*****).** Let $a \geq 4$ and $n \geq 2$. The word O^aEOEOE is not a cycle word at n.

Proof. First let $n \geq 32$, and write $z = J^a(n)$, $w = \lfloor \sqrt{z} \rfloor$, and y for the last odd letter. The suffix EOE again gives $y^3 < (n+1)^4$. The one-odd envelope on w yields $w^3 \leq 4s^2$, where s is the image after O^aEO. The last-odd cell and $n \geq 32$ upgrade this to $w < (n+1)^2$, hence $z < (w+1)^2 \leq (n+1)^4$. Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{2^{a+2}}.$$

For $n \geq 256$ and $a \geq 4$, this is the shared tail already used in Theorem 3.16.

For $2 \leq n < 256$, the cases $a = 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eoeoe` (Appendix A). For $a \geq 7$, Lemma 3.11 applies. ☐

Theorem 3.20 (mixed bunched family ***EOOEOE*****).** Let $a \geq 3$ and $n \geq 2$. The word $O^aEOOEOE$ is not a cycle word at n.

Proof. First let $a \geq 4$ and $n \geq 4$, and write $z = J^a(n)$, $u = \lfloor \sqrt{z} \rfloor$, and y for the last odd letter. The suffix EOE gives $y^3 < (n+1)^4$, hence $(y+1)^3 < 2(n+1)^4$. The two letters after u are odd, so Lemma 3.10

at length two yields $u^9 \le 2^{10}s^4 < 2^{10}(y+1)^8$. Cubing that display against the last-odd successor bound produces $u^{27} < 2^{38}(n+1)^{32}$. The same comparison as in Theorem 3.17 then gives $u < (n+1)^2$, hence $z < (u+1)^2 \le (n+1)^4$. Combined with Lemma 3.10, any such cycle word would satisfy the shared two-even tail of Theorem 3.16.

Now let $a = 3$ and $n \ge 256$. The prefix O^3 against $z < (u+1)^2$ yields $n^{27} < 2^{38}(u+1)^{16}$, and the two-odd plus last-odd geometry of the previous paragraph yields $u^{27} < 2^{38}(n+1)^{32}$. These are the same two displays as in the $a = 3$ case of Theorem 3.17, with u in place of y, and the same small/large split applies.

For $2 \le n < 256$, the cases $a = 3, 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eooeoe` (Appendix A). For $a \ge 7$, Lemma 3.11 applies. □

Theorem 3.21 (gapped leftovers as cycle words). Let $n \ge 2$. No cycle word at n has the form O^aEO^bEE with $a \ge 2$ and $b \ge 4$, or the form O^aEO^bEOE with $a \ge 2$ and $b \ge 3$.

Proof. Every cycle word has a minimum-based rotation. It is therefore enough to check that every cyclic shift of either word is an already-excluded cycle-minimum orientation.

Write w for the gapped word. In the first family, $|w| = a+b+3$. The rotation by $k = 0$ is the original word, excluded as a cycle minimum by Theorem 3.13. The rotation by $k = a+1$ is the bootstrap word O^bEEO^aE. That word has an internal even letter and last gap at least 2. If $a \ge 3$, the last-gap threshold of Lemma 3.4 at OOO and $N = 3$ excludes it. If $a = 2$, the same lemma at OO and $N = 5$ excludes every start $n \ge 5$; the remaining odd start $n = 3$ does not realize four consecutive odd letters, so it cannot follow O^b for $b \ge 4$. The rotation by $k = a+b+2$ begins with the last even letter of w, which Theorem 3.2 forbids at a cycle minimum. Every other rotation ends with an odd letter, likewise forbidden at a cycle minimum.

In the second family, $|w| = a+b+4$. The same four classes appear: the original word is Theorem 3.13; the rotation by $k = a+1$ is O^bEOEO^aE, excluded by the same last-gap thresholds, with the remaining start $n = 3$ and $a = 2$ either failing to realize four odds or, when $b = 3$, reaching 6 after $OOOE$ and then meeting an odd letter; the rotation by $k = a+b+2$ begins OE, forbidden by Theorem 3.2; and every other rotation ends odd.

The original start need not be a cycle minimum. After rotation the start is a minimum, so the hypothesis $y < n$ that blocked Theorem 3.13 does not arise. □

Length eight remains the first even-terminating expanding length outside the census. Theorem 3.12 excludes its two leftover orientations, but the census assembly of Theorem 3.8 is not extended. Length nine is the first even-terminating expanding length that admits three even letters; Theorems 3.13–3.21 treat infinite families of those leftovers, not every length-nine word. A single coarse successor power $(n+1)^K$ cannot exclude the four families of Theorems 3.17–3.20 by the same cell used for Theorems 3.15 and 3.16: at the first expanding prefix length, the exponent $K \cdot 2^a$ meets or exceeds 3^a for those four words, so those arguments use a tight last-odd cell. No exclusion of cycles of length eight or more is claimed. The census stops at length seven; no exclusion at length eight is claimed.

4. Remarks

A start $n \ge 2$ has a *descent certificate* if there exists a realized finite word w with $J^{|w|}(n) < n$.

Theorem 4.1 (uniform short certificates). Let $n \ge 2$.

1. If n is even, then the one-letter word E is a descent certificate: $J(n) = \lfloor \sqrt{n} \rfloor < n$.
2. If n is odd and $J(n)$ is even, then the two-letter word OE is a descent certificate.

Proof. For (1), $\lfloor \sqrt{n} \rfloor \le \sqrt{n} < n$ for $n \ge 2$. For (2), the word OE is realized by hypothesis, and

$$J^2(n) = \lfloor \sqrt{\lfloor n^{3/2} \rfloor} \rfloor \le n^{3/4} < n$$

for $n \ge 2$. □

The starts not covered by Theorem 4.1 are exactly the odd-to-odd starts. In particular, if $n \geq 2$ has no descent certificate of any length, then n is odd-to-odd. The converse is false: many odd-to-odd starts descend after a longer word. If every start above 1 has some descent certificate, ordinary strong induction yields arrival at 1. That hypothesis is not proved.

The first odd-to-odd image expands, so ordinary strong induction cannot fire on the complement of Theorem 4.1. A uniform run bound on expanding blocks is likewise unavailable: four consecutive expanding blocks occur already at

$$1999 \xrightarrow{OOE} 5169 \xrightarrow{OOOOEE} 50093 \xrightarrow{OOE} 193753 \xrightarrow{OOE} 887471,$$

and the relative slack of a single letter tends to 0 with the state (Section 2).

> No theorem forces every exact integer state into a contracting prefix. In particular, it is open whether every start reaches 1, and open whether a nontrivial cycle of length eight or more exists.

Lean proofs of the theorems of Sections 2–4 are in the project repository. From a clone, `lake build Problems.JugglerPaper` builds only the modules named in Appendix A.

Appendix A. Lean names

The proofs in the text are ordinary integer arguments. The following names are the corresponding Lean theorems in `formal/Problems/Juggler/`, imported by `Problems.JugglerPaper`.

Text	Lean
itinerary semantics	`follows_iff_word`, `image_eq_iterate`, `image_append`
Theorem 2.1	`image_monotone_of_follows`
Theorem 2.2	`power_bound_word`
Corollary 2.3	`power_bound_contracts`
Theorem 2.4	`global_defect_identity`
Theorem 2.5	`global_defect_eq_zero_iff_localsTight`, `global_defect_eq_zero_implies_monochrome`, `power_bound_eq_iff_extremal`
Theorem 2.6	`global_defect_append`
Corollary 2.7	`image_eq_start_defectRatio`
per-step slack	`one_plus_eta_lt_succ_sq`
Lemma 3.1	`odd_cell_unique`
Theorem 3.2	`cycle_word_formally_expanding`, `cycleMin_start_odd`, `cycleMax_start_even`, `cycleMin_not_end_odd`, `square_scale_superquadratic`, `cycleMin_to_even_superquadratic`
Lemma 3.3	`lower_growth_word`
Lemma 3.4	`oo_suffix_threshold`, `ooo_suffix_threshold`, `threshold_inherits_odd_append`, `cycle_last_even_interval`, `no_cycle_odd_run_append_even`
Lemma 3.5	`no_cycle_word_oooeoe`, `no_cycle_word_ooooee`
Theorem 3.6	`no_cycle_word_length_le_six`
Lemma 3.7	`no_cycle_word_ooooeoe`, `no_cycle_word_oooooee`
Theorem 3.8	`no_cycle_word_length_le_seven`, with `no_cycle_word_ooeoooe`, `no_cycle_word_oooeooe`
Lemma 3.9	`cycle_trailing_evens_lt`
Lemma 3.10	`lowerDenom_replicate_odd`, `odd_run_lower_growth`

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Lemma 3.11	`no_follows_seven_odds_of_lt256`
Theorem 3.12	`no_cycle_word_two_even_ee`, `no_cycle_word_two_even_eoe`
Theorem 3.13	`no_cycleMin_gapped_three_even_ee`, `no_cycleMin_gapped_three_even_eoe`
Theorem 3.14	`no_cycle_word_three_even_eee`, with `no_cycle_word_ooooooeee`
Theorem 3.15	`no_cycle_word_three_even_eoee`
Theorem 3.16	`no_cycle_word_three_even_eooee`
Theorem 3.17	`no_cycle_word_three_even_eoooee`
Theorem 3.18	`no_cycle_word_three_even_eeoe`
Theorem 3.19	`no_cycle_word_three_even_eoeoe`
Theorem 3.20	`no_cycle_word_three_even_eooeoe`
Theorem 3.21	`no_cycle_word_gapped_three_even_ee`, `no_cycle_word_gapped_three_even_eoe`
Theorem 4.1	`even_finiteProgress`, `odd_even_finiteProgress`
no certificate ⇒ odd-to-odd	`no_finiteProgress_implies_odd_odd`
induction to 1	`reachesOne_of_all_finiteProgress`
four-block chain	`four_block_pe_1999`

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